

L02 Reflective Journal_FA26_ML_6171_163474

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Elquin Ponce

This lab gave me a hands-on look at how the core Python tools used in machine learning actually work together. I had some prior exposure to Python, but I hadn't used pandas, NumPy, and matplotlib together on the same dataset before, so seeing each library's role side by side made the workflow click in a way that reading about it separately didn't.

Working with the Iris dataset made the process concrete. Once the data was loaded into a pandas DataFrame, it was easy to see all 150 samples across the four measurements and three species, but scanning the raw numbers alone didn't reveal much. Using NumPy to calculate the mean and standard deviation of sepal length was a useful reminder that basic statistics can highlight patterns before any visualization or modeling happens.

The scatter plot was the part that made the biggest difference for me. Seeing sepal length plotted against sepal width, colored by species, made the separation between setosa and the other two species immediately obvious in a way the table of numbers never did. Versicolor and virginica overlapped noticeably, which told me that sepal measurements alone probably aren't enough to reliably tell those two apart - something I wouldn't have picked up on without visualizing it first.

This lab also shifted how I think about the start of a machine learning project. I tend to think of ML as training a model and getting a prediction, but this exercise made it clear that understanding the shape and structure of the data comes first. Exploring the data, checking basic statistics, and visualizing relationships all inform decisions that get made before any actual model is built.

Going forward, I'm curious whether plotting petal length and petal width instead of the sepal measurements would separate versicolor and virginica more clearly, since the species means showed a bigger gap in petal measurements than in sepal width. Overall, this lab helped me get more comfortable treating data exploration as its own step, rather than something to rush through on the way to building a model.

Aaliyah Phillips

Before going into this assignment, I was not familiar with Google Colab. I do have experience in using VSCode and R, so I think that I was easily able to transfer some of my knowledge from those platforms to this one easily. I appreciate how much simpler and beginner-friendly Colab is, I was able to just jump into creating cells and running codes. Installing Python libraries was straight forward and easy, had no issues with them whatsoever.

After loading up the Iris dataset, I tried spending some time comprehending the dataset before moving on. While the Pandas data frame did a bit to make the data readable to me, I still had a hard time really understanding what it was trying to convey. I took this as a sign that I need to spend some time practicing how to understand pandas indexing, It wasn't until I moved on to the visualization aspect of the lab that it started to make sense to me. The scatterplot helped me see the differences in length and width between the species. For example, I could tell easily that the average width of the setosa was greater compared to the other two. Next came the Basic operations; finding the mean values for each feature help solidify the differences in size, length, and width of the different species . Lastly, we created a simple bar chart to showcase the total number of species counts. Compared to the scatterplot, I don't think this chart was as effective or necessary, as all it showed that the sample count was even (50 for each species = 150 samples in total).

This lab has showed me that firstly, we have the “clean up” data sets before we can move onto anything involving graphing and calculations. Converting the dataset to pandas data frame made the process of understanding the information a little easier, though I was still uncertain of how to read it. After completing this lab, I also realized that I should probably brush up more on my knowledge regarding Python and R. While using Colab, I was reminded that the tools here were identical to R specially, but also shared similarities with Python as far as coding. So I plan to study those subjects again as I progress into this course. I believe it will assist me as I learn more about machine learning.

Diane Lenhoff

This lab helped me better understand how the different tools in Python work together. I have used Google Colab and Python before, so that part was somewhat familiar. What was new for me was seeing how Pandas, NumPy, and Matplotlib could each be used differently with the same dataset. Instead of just running the code to see if it worked, I started paying more attention to what the results were actually showing me.

When I first looked at the rows and numbers in the Iris dataset, I really couldn't see any clear patterns. Pandas helped organize the data and made it easier to understand. I also used NumPy to calculate the mean and standard deviation for sepal length, which showed me how even simple statistics can help make sense of the data.

The scatter plot was probably the most helpful part for me. Once I saw the data on a graph, the differences between the flowers made a lot more sense. Setosa stood out more clearly, while versicolor and virginica overlapped more. I could also see that setosa generally had shorter but wider sepals. Seeing the graph made this much easier to understand than just looking at the numbers.

One thing I learned from this lab is that there is a lot to do before training a machine learning model. Before this, I mostly thought about machine learning as training a model and getting a prediction. Now I understand why you need to explore the data and look for patterns first. It also made me curious whether using petal length and petal width would separate the three flower types even more clearly.

Overall, I feel more comfortable looking at a dataset and thinking about what the information means instead of only worrying about whether my code runs correctly. My biggest takeaway was how much easier it is to recognize patterns when the data is visualized. I think taking the time to understand the data first will help me make better decisions as I learn more about machine learning.

Seth Alvarez

I noticed that the red dots that represented setosa were isolated from the other two species when I ran the scatter plot. Reading the rows one at a time would have required me to hold 150 pairs of numbers in my head to notice that one group clustered apart from others. The plot showed all of them at once, which made the gap obvious. The numbers showed that setosa had the highest value for width and the lowest for length compared to the other species, confirming what I saw in the plot. Now that I've seen the plot and the numbers agree, I would use both on the next dataset rather than relying on one alone.

The petal measurements spread the three species far apart, while the sepal measurements clustered them closer together. The plot used measurements that separated the species least, explaining why versicolor and virginica overlapped so heavily. This left me wondering whether plotting the petal length against petal width would separate all three species clearly, given how far apart their mean petal values were. I realize that the features you choose to plot from a dataset change what conclusion is reached, and a poor choice can make a solvable problem look unsolvable.